

Notes: Milgrom & Roberts (JPE, 1986)

Price and Advertising Signals of Product Quality

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1 Big picture

- **Question.** Can directly uninformative advertising signal the quality of an experience good when the firm can also choose the introductory price?
- **Setting.** A monopolist introduces a new experience good whose quality is privately known by the firm but not by consumers. Consumers observe the introductory price and advertising outlay before buying, learn through use or communication, and may buy again later.
- **Central mechanism.** Repeat purchases make initial sales more valuable to high-quality firms, but once price is a signal, the marginal value of attracting an initial sale is not enough to explain advertising by itself. Price and advertising are jointly chosen signals.
- **Main result.** A high-quality firm may use both price and dissipative advertising to separate itself from a low-quality firm. Advertising occurs only when price alone cannot cheaply create enough separation.
- **Equilibrium refinement.** The paper does not accept all sequential equilibria. It refines equilibria using sequential elimination of dominated strategies and a Kreps-style intuitive criterion, which sharply reduces implausible pooling and separating equilibria.
- **Policy insight.** Banning purely dissipative advertising need not improve welfare. If advertising is banned, a high-quality firm may signal through a more distorted price, which can leave consumers with the same information but worse prices.
- **Economic takeaway.** Wasteful-looking advertising can be useful because it allows high-quality producers to signal at lower real distortion than price alone. The informative content is not the ad message but the costliness and equilibrium incentive compatibility of the signal.

2 Incremental contribution of the paper

- **Formalizes Nelson’s intuition with pricing included.** Nelson argued that uninformative advertising can signal product quality because repeat purchases make initial sales more valuable for high-quality sellers. Milgrom and Roberts show that this intuition is incomplete unless the introductory price is also modeled as a choice variable and potential signal.
- **Moves from one-dimensional to multidimensional signaling.** Earlier signaling models often used a single signal. This paper studies two simultaneous signals—price and advertising—for one hidden characteristic, product quality.
- **Explains when advertising is used in addition to price.** The paper identifies cases where price alone separates types and cases where advertising is added because the least-cost separating signal uses both instruments.
- **Shows that advertising may be welfare improving despite being dissipative.** If advertising is an efficient signaling dimension, banning it may force firms to signal through distorted prices and can be Pareto worsening.
- **Adds equilibrium-selection discipline.** The paper illustrates how dominated-strategy reasoning and the Kreps criterion can eliminate many implausible equilibria in a multi-signal game, yielding a small set and sometimes uniqueness.
- **Corrects a common intuitive explanation.** The paper shows that the phrase “high-quality firms value an initial sale more” is not well defined once price and beliefs are endogenous. The marginal value of sales depends on the way the sale is generated: price, advertising, or direct quantity shifts.

3 Model environment and primitives

3.1 Experience goods and two quality types

A firm introduces a new product of either high quality H or low quality L , with $H > L$. The firm observes its own quality. Consumers do not observe quality before the first purchase. Product quality is exogenous: the firm has already developed the product and now chooses how to introduce it.

The analysis applies to *experience goods*. Consumers cannot verify quality before use, but after buying they learn from direct experience or communication with other users. Thus repeat sales are crucial.

Interpretation: The model is not about moral hazard in quality provision. It is about adverse selection: the firm already knows quality, consumers do not, and the firm chooses observable signals before consumers learn.

3.2 Signals: introductory price and dissipative advertising

The firm chooses:

- an introductory price P ;
- an advertising expenditure A beyond purely informative advertising.

Advertising is modeled as directly uninformative and purely dissipative. It does not directly shift demand or production costs. Its only possible role is to change beliefs.

$$p(P, A) = \Pr(q = H \mid P, A)$$

denotes consumers' posterior belief that the firm is high quality after observing price and advertising.

Interpretation: This is why the paper is powerful: it studies advertising that seems wasteful in a direct-demand sense, and asks whether equilibrium incentives can make that waste socially meaningful.

3.3 Profit notation

Let

$$\Pi(P, q, p) - A$$

be the expected present value of profits for a firm of true type $q \in \{L, H\}$ that sets introductory price P , spends A , and is believed to be high quality with probability p .

When customers are certain the firm is type $Q \in \{L, H\}$, define

$$\pi(P, q, Q) = \Pi(P, q, p_Q),$$

where $p_H = 1$ and $p_L = 0$. Thus:

- $\pi(P, H, H)$ is profit of a true high-quality firm perceived as high quality;
- $\pi(P, L, H)$ is profit of a true low-quality firm mimicking high quality;
- $\pi(P, L, L)$ is profit of a low-quality firm correctly perceived as low quality.

Let P_H^H and P_L^L denote full-information optimal prices for high and low quality, respectively.

4 Equilibrium concepts and specifications

4.1 Separating equilibrium: initial condition

A separating equilibrium requires a choice (P, A) that the high-quality firm wants to make, but the low-quality firm does not want to mimic. Proposition 1 states that a separating sequential equilibrium exists if and only if for some (P, A) ,

$$\pi(P, H, H) - \pi(P_H^L, H, L) \geq A \geq \pi(P, L, H) - \pi(P_L^L, L, L). \quad (1)$$

In any separating equilibrium, the low-quality firm chooses its full-information point $(P_L^L, 0)$, and the high-quality firm chooses some (P, A) satisfying the incentive-compatibility inequalities.

Interpretation: The right inequality prevents low quality from mimicking high quality. The left inequality ensures high quality is willing to bear the signaling cost rather than be perceived as low quality.

4.2 Dominated-strategy refinement and the high-quality firm's problem

The paper then eliminates equilibria supported only by implausible off-path beliefs. After sequential elimination of dominated strategies, the high-quality firm solves:

$$\max_{P, A} \pi(P, H, H) - A \quad \text{s.t.} \quad \pi(P, L, H) - A \leq \pi(P_L^L, L, L), \quad P, A \geq 0. \quad (2)$$

If the optimum has positive advertising, the price component solves:

$$\max_P \pi(P, H, H) - \pi(P, L, H) \quad \text{s.t.} \quad \pi(P, L, H) - \pi(P_L^L, L, L) > 0. \quad (3)$$

Interpretation: Advertising appears only when the high-quality firm has reached the least-cost price point for separating and still needs extra costly differentiation. In that case, advertising fills the remaining incentive-compatibility gap.

4.3 When does advertising occur? Conditions (4)-(7)

A necessary condition for advertising to occur is that the high-quality firm's full-information price does not itself separate the type:

$$\pi(P_H^H, L, H) > \pi(P_L^L, L, L). \quad (4)$$

This says that a low-quality firm would be willing to choose the high-quality full-information price if doing so caused consumers to believe it was high quality.

A further condition for an interior advertising solution is that there exists a tangency price P^T satisfying:

$$\pi(P^T, L, H) > \pi(P_L^L, L, L). \quad (5)$$

The curvature of the difference in high-type and low-type profits determines whether the tangency is a maximum or a minimum:

$$\pi(P, H, H) - \pi(P, L, H) \text{ is pseudoconcave in } P, \quad (6)$$

$$\pi(P, H, H) - \pi(P, L, H) \text{ is strictly pseudoconvex in } P. \quad (7)$$

Proposition 3: if (4)–(6) hold, there is a separating equilibrium with positive advertising. If (6) is strict, the separating equilibrium is unique. If (4) or (5) fails, or if (7) holds, all separating equilibria have zero advertising.

Interpretation: Conditions (4)–(7) answer the central economic question: price alone separates types unless the low-quality firm would mimic the high-quality price and unless price distortions become too expensive for high quality before separation is achieved.

4.4 Kreps criterion and multidimensional single-crossing

The paper uses a Kreps-style intuitive criterion to eliminate many pooling equilibria. The analog of single crossing in this multi-signal setting is condition (8): at every relevant (P, A, p) , there is a feasible direction in price-advertising-belief space that would benefit the high-quality type but not the low-quality type.

Informally, condition (8) says: for any off-path point, if a deviation could only be profitable for H and not for L , consumers should infer that the deviator is high quality.

Proposition 4: if condition (8) holds, a separating equilibrium exists and satisfies the Kreps criterion, and no pooling equilibrium satisfies the criterion.

Interpretation: This is the paper’s methodological contribution. It shows how equilibrium refinements can discipline a multidimensional signaling game where Bayes’ rule alone leaves too many off-path beliefs.

5 Explicit repeat-purchase example

5.1 Customers, satisfaction, and demand

Quality is the probability that a randomly selected consumer finds the product satisfactory. Consumers are uniformly distributed on $[0, R]$. If a consumer with valuation r buys at price P , utility is $r - P$ if the good is satisfactory and $-P$ otherwise.

If the posterior probability of high quality is ρ , expected quality is

$$Q(\rho) = \rho H + (1 - \rho)L.$$

The marginal initial buyer satisfies

$$Q(\rho)r^* = P, \quad \text{so} \quad r^* = \frac{P}{Q(\rho)}.$$

Initial demand is therefore

$$R - \frac{P}{Q(\rho)}.$$

Interpretation: Uncertainty lowers willingness to pay. Consumers pay for expected quality, not true quality. After experience, only satisfied buyers return.

5.2 Repeat demand and post-entry pricing

A fraction q of initial buyers are satisfied and remain candidates for repeat purchase. Post-entry demand is based on the satisfied customer base. The firm sets a post-entry price weakly above the initial marginal valuation r^* , so the first-period marginal consumer earns no surplus on repeat purchases.

The model obtains profit functions corresponding exactly to the earlier reduced-form notation $\pi(P, q, H)$ and $\pi(P, q, L)$.

Interpretation: The example formally verifies the repeat-sales mechanism. High quality creates more satisfied buyers, but the value of those buyers depends on introductory price, beliefs, and post-entry pricing.

5.3 Cost comparative statics

Let C_H and C_L denote the costs of high and low quality. The example generates clear comparative statics:

- If $C_H < C_L$, high quality signals by cutting price or giving away samples; no advertising is used.
- If C_H is slightly above C_L , the unique separating equilibrium uses positive advertising and a price adjustment.
- If C_H is much larger than C_L , price alone again separates types through a high introductory price; advertising disappears.
- At $C_H = C_L$, there is an interval of separating price-advertising choices.

Interpretation: Advertising is most likely when high and low quality are sufficiently similar in cost that price alone cannot create cheap separation. When cost differences are large, either low prices or high prices can carry the signal without dissipative advertising.

6 Identification and theoretical strategy

6.1 No empirical identification

This is a theory paper. There is no dataset, treatment, or estimation design. The relevant identification is theoretical: the paper identifies incentive conditions under which price and advertising can credibly separate product quality.

6.2 Step 1: Separate direct information from signaling

The paper begins with advertising that does not directly inform consumers about quality. The advertisement's only possible role is to change beliefs by being costly in an incentive-compatible way.

6.3 Step 2: Add price as a simultaneous signal

The major modeling move is to treat both price and advertising as choice variables. This is crucial because once price is allowed to signal, advertising is not automatically needed.

6.4 Step 3: Use repeat purchases to create type-dependent payoffs

The value of an initial sale depends on future repeat purchases. High quality generates more satisfied customers and more repeat demand. But the marginal value of attracting an initial buyer depends on whether the buyer is attracted by lowering price, raising advertising, or changing beliefs.

6.5 Step 4: Refine equilibrium beliefs

The paper starts from sequential equilibrium and then imposes dominated-strategy reasoning and the Kreps criterion. This eliminates many equilibria supported by unreasonable beliefs about off-path price-advertising choices.

6.6 Step 5: Characterize when signals are used

The key theoretical results identify regions in which:

- price alone signals high quality;
- price plus advertising signals high quality;
- advertising would be redundant;
- pooling equilibria survive or fail refinement.

Interpretation: The paper is not trying to “measure” advertising effects. It is identifying a mechanism and characterizing its equilibrium implications.

7 Figures and detailed interpretations

7.1 Figure 1: Existence of separating equilibrium

Read: Figure 1 plots introductory price P horizontally and advertising A vertically. The two isoprofit curves represent the low-quality firm mimicking high quality and the high-quality firm separating. The shaded region contains price-advertising pairs satisfying the incentive inequalities for separation.

Interpretation: Points in the shaded region can support separating sequential equilibria before refinement. The diagram also shows why many equilibria are implausible: some off-path choices would be dominated for low quality, so consumers should not treat them as evidence of low quality.

Economic message: Separation is an incentive-compatibility problem. The high-quality firm must spend enough, through price distortion and/or advertising, to make imitation unattractive for low quality.

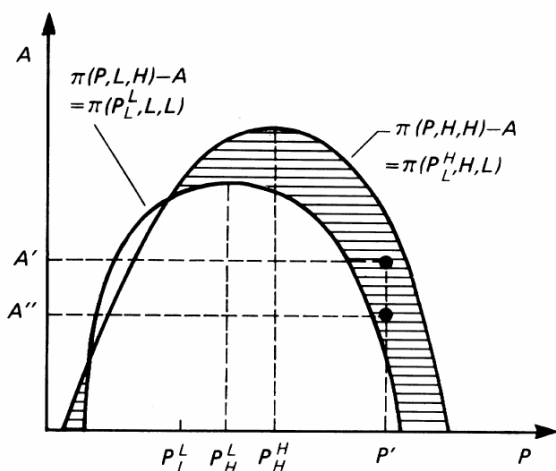


FIG. 1.—Existence of separating equilibrium

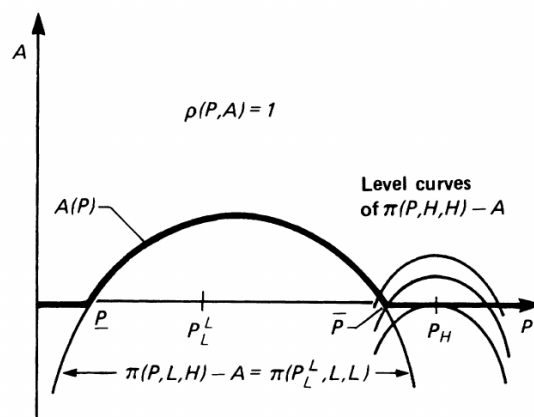


FIG. 2.—Condition (4) fails; equilibrium at the full-information price

7.2 Figure 2: Condition (4) fails; equilibrium at the full-information price

Read: Figure 2 shows a case where the high-quality full-information price already lies outside the region where a low-quality firm would mimic. Therefore, price alone conveys quality and advertising is unnecessary.

Interpretation: The figure covers two economic cases. If high quality is very costly and targeted to a limited market, a high introductory price may separate. If high quality is very cheap and intended for a mass market, a low introductory price or free sample may separate.

Economic message: Advertising is not used when the high-quality firm's natural price is already too unattractive for low quality to mimic.

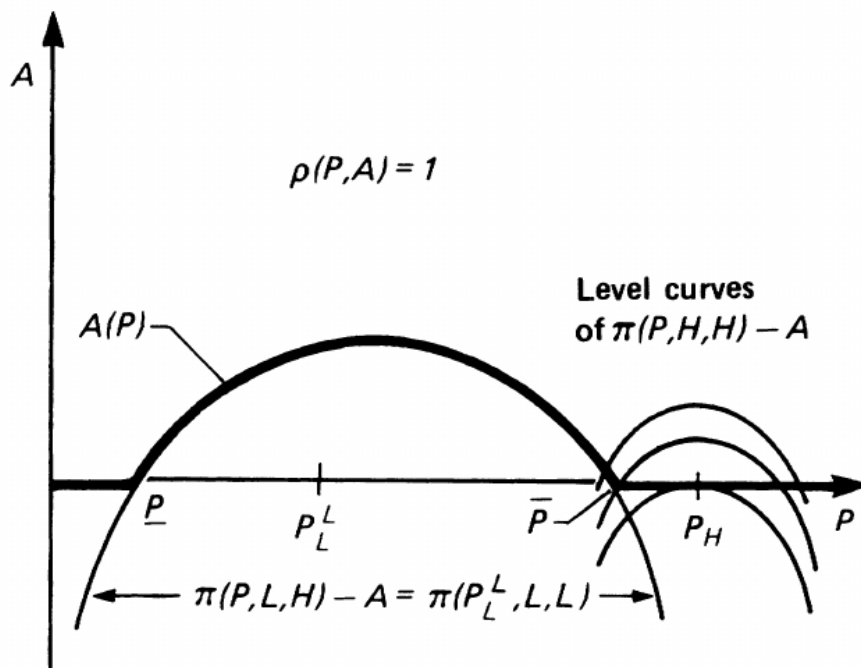


FIG. 2.—Condition (4) fails; equilibrium at the full-information price

Figure 2: Condition (4) fails; equilibrium at the full-information price.

7.3 Figures 3 and 4: Zero-advertising boundary cases

Read: Figure 3 shows the case where condition (5) fails. Price adjustments toward P^T remain cheaper for high quality than for low quality, so price alone is used and advertising is zero. Figure 4 shows the case where condition (7) holds, so the tangency is a minimum rather than a maximum and the solution is again a boundary point with zero advertising.

Interpretation: These figures show that the mere possibility of price signaling is often enough. Advertising appears only if the least-cost price distortion is insufficient and an interior tangency exists.

Economic message: Price is a powerful signal because it directly changes demand and repeat-sales base. Advertising is used only when price distortions become relatively expensive.

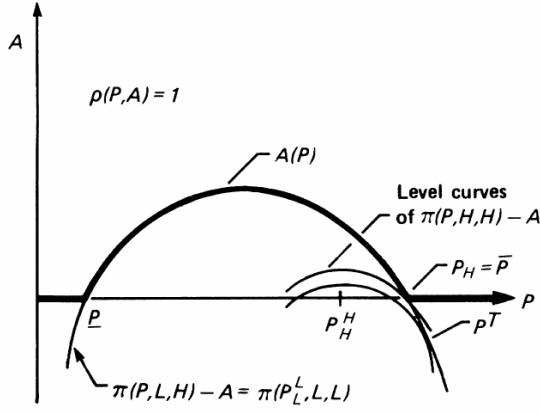
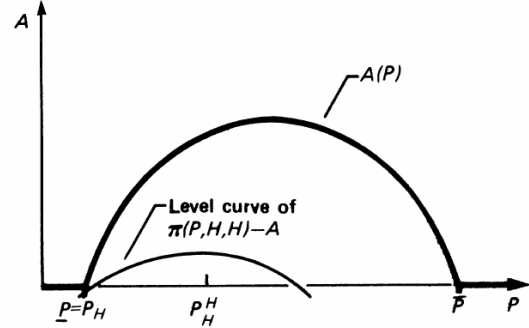


FIG. 3.—Condition (5) fails; equilibrium at $(\bar{P}, 0)$

(a) Figure 3: condition (5) fails.

$\bar{P}, \bar{P}]$ (see fig. 4).

Figures 5 and 6 illustrate possible cases with positive advertising.



(b) Figure 4: condition (7) holds.

7.4 Figures 5 and 6: Positive advertising with lowered or raised price

Read: Figure 5 shows an equilibrium with positive advertising and a lowered introductory price. Figure 6 shows an equilibrium with positive advertising and a raised introductory price. In both, advertising and price jointly carry the signal.

Interpretation: The sign of the price distortion is not fixed. A high-quality firm may signal by charging less or more than the full-information high-quality price, depending on relative cost conditions and demand/repeat-sales effects.

Economic message: The model predicts that high-quality launch strategies can look very different: premium pricing plus big advertising, or low introductory pricing plus big advertising. Both can be rational signals.

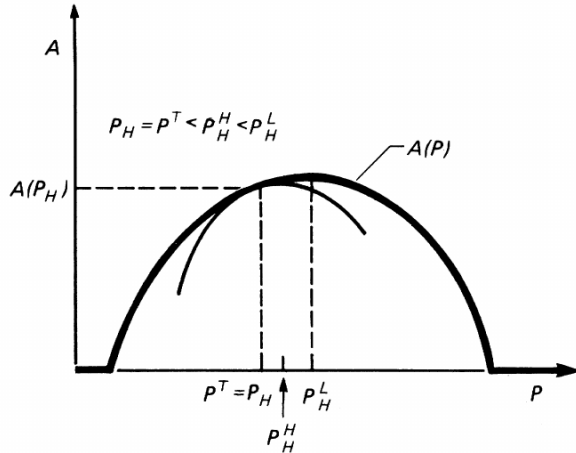


FIG. 5.—Equilibrium with positive advertising and lowered price ($P_H < P_H^H$)

(a) Figure 5: positive advertising and lowered price.

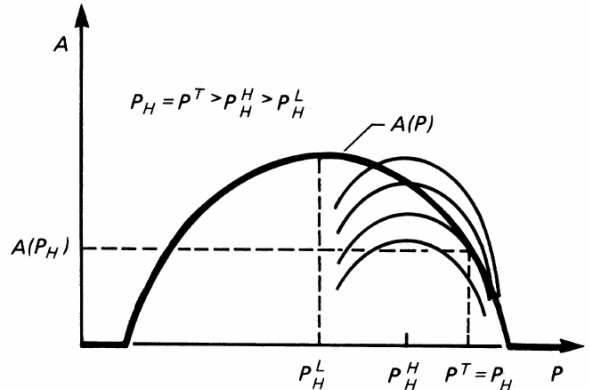


FIG. 6.—Equilibrium with positive advertising and raised price ($P_H > P_H^H$)

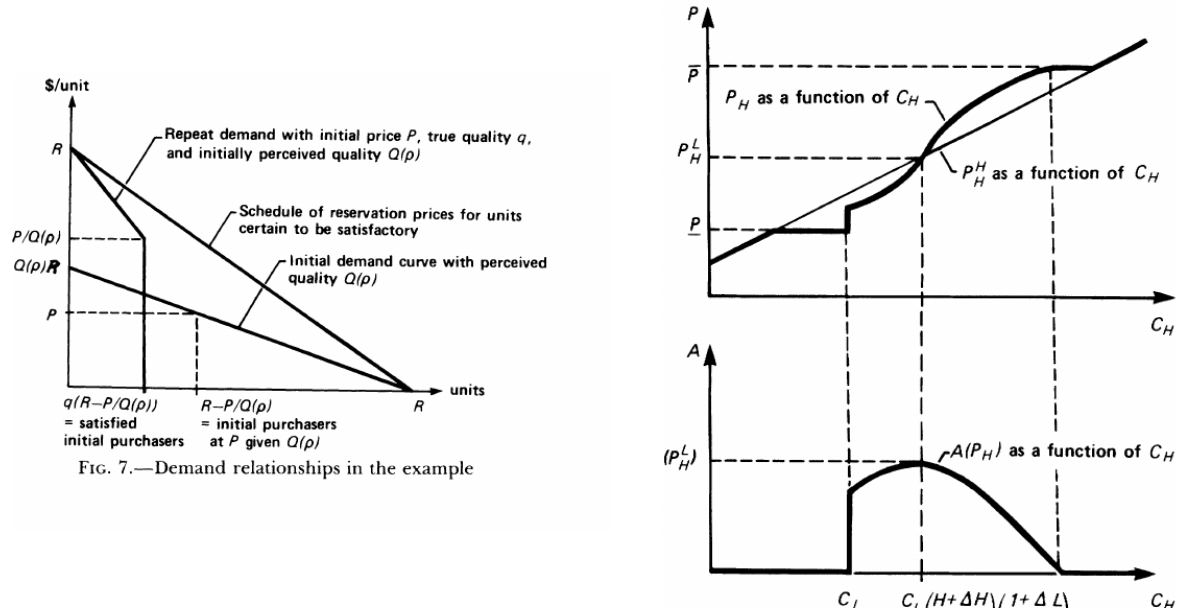
(b) Figure 6: positive advertising and raised price.

7.5 Figure 7: Demand relationships in the explicit example

Read: Figure 7 decomposes initial demand, repeat demand, and the reservation-price schedule. The marginal initial purchaser satisfies $r^* = P/Q(\rho)$, where $Q(\rho)$ is expected quality.

Interpretation: This figure explains why repeat purchases are central. A high-quality product generates more satisfied repeat buyers, but the size and profitability of that repeat base depend on the introductory price and consumer beliefs.

Economic message: Initial price affects not only current revenue but the composition and size of the future repeat-customer base.



7.6 Figure 8: Equilibrium values of advertising and introductory price as C_H varies

Read: Figure 8 plots the high-quality firm's equilibrium introductory price and advertising as the high-quality cost C_H changes. Advertising is zero when high quality is sufficiently cheap or sufficiently costly relative to low quality, and positive only in an intermediate range.

Interpretation: The figure summarizes the paper's main comparative statics. If high quality is cheap, a low price signals quality. If high quality is costly, a high price signals quality. Advertising is needed only when cost differences are moderate enough that price alone is not the cheapest separating signal.

Economic message: The model predicts a nonmonotonic relation between product-quality cost differences and dissipative advertising. The most intense uninformative advertising occurs neither when quality is very cheap nor when it is very expensive, but in an intermediate region.

8 Interpretive synthesis

- **Why advertising can be informative without content.** Advertising is useful because it is costly in a way that low-quality firms may not want to mimic once future repeat purchases and price choices are accounted for.
- **Why price matters.** Price is not just a revenue instrument. It changes quantity sold, the set of marginal buyers, the repeat-purchase base, and the credibility of a quality claim.

- **Why both signals may be used.** If one signal alone is a more expensive way to deter mimics, the high-quality firm spreads the signaling burden across price and advertising.
- **Why bans can backfire.** If advertising is banned, firms may use a more distortive price to communicate the same information, possibly worsening welfare.
- **Why equilibrium refinement is essential.** Without restrictions on off-path beliefs, the model has many equilibria. The economic conclusions require eliminating equilibria sustained by implausible beliefs about dominated deviations.

9 Short summary

- The paper studies introductory price and uninformative advertising as signals of experience-good quality.
- Consumers cannot observe quality before purchase, but learn through experience and repeat purchases.
- A high-quality firm may use price, advertising, or both to separate itself from a low-quality firm.
- Advertising is used only when price alone cannot cheaply deter mimicry.
- The sign of the price distortion is ambiguous: high quality may signal through a low price or a high price.
- The explicit repeat-purchase model shows how cost differences determine whether advertising is positive.
- The paper uses sequential equilibrium, dominated-strategy reasoning, and a Kreps-style intuitive criterion to refine the equilibrium set.
- Banning dissipative advertising can be welfare reducing because firms may substitute toward more distorted prices.

10 Conclusion

10.1 Core conclusion

Milgrom and Roberts formalize the idea that uninformative advertising can signal product quality, but they show that this logic must be analyzed jointly with introductory pricing. Advertising is not automatically the natural signal of quality; price is often sufficient. Advertising appears when it is the least costly additional signal needed to prevent low-quality firms from mimicking.

10.2 Economic intuition

For a new experience good, consumers learn only after initial purchase. A high-quality firm benefits from a large satisfied repeat customer base, but the value of initial sales depends on the introductory price and consumer beliefs. Price and advertising therefore jointly shape the credibility and cost of signaling. The high-quality firm chooses the least-cost combination that makes imitation unattractive for low quality.

10.3 Incremental contribution in one sentence

The paper's distinctive contribution is to transform Nelson's informal repeat-purchase advertising story into a formal multi-signal equilibrium model in which price and advertising are both strategic signals and equilibrium refinements determine which signaling outcomes are credible.

10.4 Final takeaway

Seemingly wasteful advertising can be economically useful because it may reduce more harmful price distortions. The informational role of advertising cannot be understood without jointly modeling prices, repeat purchases, and equilibrium beliefs.